Review of the paper "Bayesian Estimation and Forecasting of Internal Migration Rates: An Application to the UK"

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Overview of the Paper

The paper proposes a Bayesian hierarchical model for estimating and forecasting internal migration rates by age and sex across different regions of the United Kingdom. I believe that the proposed framework has considerable potential and could be fruitfully applied to similar analyses in other national contexts. In addition, the mathematical structure of the model is described in considerable detail and is generally presented in a clear and accessible manner.

Nonetheless, I believe that the manuscript would benefit from revision before publication. My main comments concern the need to strengthen the model validation analysis, improve the graphical presentation of some figures, add relevant references to the introduction, provide a more extensive discussion of the substantive and methodological implications of the results in the conclusion, and correct a number of issues in the bibliography. Detailed comments are provided in the following section.

Broadly speaking, I consider the manuscript to represent a promising contribution to the fields of Bayesian demography and applied statistics. If the authors adequately address the comments raised below and incorporate the suggested revisions where appropriate, I believe the manuscript would be suitable for publication in *Data Analytics and Topology*.

Detailed Comments

1. The introduction is generally well written, and the authors have done an excellent job of motivating the use of Bayesian modeling for the estimation and forecasting of internal migration rates. Nonetheless, I believe that the discussion of previous work on subnational demographic estimation and forecasting could be better supported by references, particularly in lines 3–4 on page 2. In this regard, I would suggest that the authors consider citing the following studies:

- Alexander, M., & Alkema, L. (2022). A Bayesian cohort component projection model to estimate women of reproductive age at the subnational level in data-sparse settings. *Demography*, 59(5), 1713-1737. doi: https://doi.org/10.1215/00703370-10216406 (link)
- Omenti, R., Alexander, M., & Barban, N. (2026). A Bayesian Model to Estimate Male and Female Fertility Patterns at a Subnational Level. doi: https://osf.io/preprints/socarxiv/kshr7$_v$2 (link)

2. In the last two lines of the introduction paragraph on page 2, the link to the GitHub repository is missing. I suggest the authors to fix this.

3. When presenting the main geographical areas used to compute the internal migration rates, I think it would be useful to include a map highlighting the location of the regions considered in the analysis. This would help readers who are less familiar with the geography of the country to better contextualize the spatial patterns discussed in the manuscript.

4. I did not understand the rationale behind considering the period $2012 - 2024$. Is this choice driven by data availability, i.e., because information on internal migration flows by age and sex only became available from 2012 onward? Or are there some other reasons for choosing this specific time window? I would suggest clarifying this aspect in the "Data" section.

5. I would suggest reconsidering the background used in the figures. Rather than the current gray background, I would personally opt for a white background, which I find cleaner and more visually appealing and which may also improve readability.

6. When discussing Figure 3, I was particularly struck by the increase in migration flows from London to the East of England and the South East during the COVID-19 pandemic. What I find especially interesting is that, particularly for the South East, migration rates from London appear to have remained above their pre-pandemic levels even after the initial shock. Is there a possible explanation for this persistence? For example, it could reflect more structural changes in residential preferences, commuting patterns, or the diffusion of remote and hybrid work. I think it would be useful to add a brief discussion of this pattern in the data section.

7. When presenting the main data model in Equation 1, I would avoid denoting the population exposure term, which is non-stochastic, by w. I would suggest using a more intuitive notation, such as P, to make its interpretation immediately clear to the readers.

8. I find the use of the symbol τ to denote the variances of the prior distributions somewhat confusing. In the Bayesian literature, the Greek letter τ is commonly used to denote a precision parameter, i.e., the inverse of the variance. To avoid potential ambiguity, I would suggest using σ^2 to denote the prior variances.

9. I believe that the model-checking analysis has been conducted very rigorously, as has the comparison of the alternative model specifications using the well-established WAIC criterion. However, I think this section could be further improved by including an out-of-sample validation exercise. One possible approach would be to leave out the most recent observation period, for example 2022–2024, fit the models using the preceding data, and assess how closely the resulting forecasts align with the subsequently observed migration rates. Forecast accuracy could be evaluated using standard metrics, such as the mean absolute error or mean absolute relative error. In addition, I would recommend assessing the empirical coverage of the posterior predictive intervals. This would provide useful information on whether the models are not only accurate in terms of point predictions, but also appropriately calibrated.

10. In Figures 4-6, the meaning of the point estimates is not entirely clear. I assume that they represent either the posterior median or posterior mean of the estimated internal migration rates, but this should be stated explicitly in the text and figure captions.

11. Again with reference to Figures 4-6, I would recommend displaying the 95% credible intervals not only for the forecast period, but also for the observed years 2014, 2018, and 2022. This would provide a more complete representation of the uncertainty surrounding the estimates.

12. I believe that presenting three origin–destination combinations in the main manuscript is more than sufficient. However, I would recommend reporting the results for all remaining origin–destination combinations in the appendix. This would allow interested readers to assess the robustness and generalizability of the findings across the full set of migration flows without overloading the main text.

13. I believe that the conclusion section is currently too brief and would benefit from a more nuanced discussion of the significance and implications of the proposed methods and empirical findings. In particular, the authors could further emphasize the potential extensions of the methodological framework and discuss other settings or research questions to which the proposed approach could be applied. At the same time, I would encourage the authors to provide a more explicit discussion of the limitations of the proposed Bayesian models, including the conditions under which their performance or applicability may be constrained. Such additions would help readers better assess both the contribution and the broader applicability of the proposed framework.

14. The bibliography contains several typographical and formatting inconsistencies, particularly with respect to the use of lower- and upper-case letters. These issues may originate from the formatting of the .bib file.

I would therefore suggest that the authors carefully review the bibliography entries and correct any formatting errors, including missing capitalization and inconsistencies in the reported bibliographic information, such as omitted article numbers, issue numbers, or volume numbers where available.